

RESEARCH ARTICLE

Evaluation of System Performance for Load Dispatch Feasibility Under N-1 Generator Contingencies in Day-Ahead Unit Commitment

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ABSTRACT Day Ahead (DA) Unit Commitment (UC) enhances system preparedness for real-time dispatch. N-1 thermal generator contingencies challenge the reliability of DA UC. This paper proposes a methodology to evaluate the impact of individual generator outages on UC operational cost, reliability, and operating margin of the scheduled DA dispatch plan, aiding better real-time dispatch decisions. System performance is assessed using criticality, robustness, reliability, and operating margin, with the Contingency Margin (CM) as an implementation indicator. The approach identifies robust and weak buses to determine system robustness and criticality. The paper addresses system reliability during contingencies via the Loss Of Load Probability (LOLP) index. Analysis includes three case studies: Base Case, Case 1, and Case 2. The Base Case executes DA UC without n-1 contingencies. Case 1 analyzes n-1 thermal generator contingency-based DA UC with a CM corresponding to a 10% spinning reserve. Weak and robust buses are identified in Case 1. In Case 2(a), 2(b) and 2(c), CM is evaluated with a gradual reduction in spinning reserve from 10% to 8%, then to 5%, and finally to 0%, while executing n-1 generator contingencies for DA UC. System performance is assessed using the proposed parameters and methodology. The Dynamic Programming technique optimizes DA UC, analyzed on the 24-bus, 26-generator IEEE Reliability Test System. Across all cases, DA UC costs range from \$774,020 to \$895,400 under varying contingencies. Critical generators (IDs 6, 7) and weak buses (Buses 18, 21) are consistently identified. CM improves from 0 MW in Case 2(c) to 310.5 MW in Case 1; LOLP reduces from 0.0501 to 0; and the operating margin increases from -0.0001 to 0.05. Results confirm that the proposed methodology effectively identifies system vulnerabilities and quantifies the impact of reserve levels on dispatch feasibility, emphasizing the importance of incorporating contingency-aware planning in DA UC and reserve allocation strategies. It provides a practical tool for operators to enhance DA UC planning, reserve allocation, and grid resilience under contingency conditions. This work provides insights for system operators and market planners to estimate performance parameters for DA UC, maintaining grid resilience against unexpected generator outages during real-time load dispatch.

INDEX TERMS N-1 generator contingency, operating margin, system reliability, system criticality, system robustness, thermal generation, unit commitment.

I. INTRODUCTION

Contingency analysis entails the examination of the implementation of outages for transmission lines and generators, whether individually or simultaneously. An outage

experienced by a single entity at one time is known as an n-1 contingency. When two entities experience outages concurrently, it is called an n-2 contingency, also known as a simultaneous outage. Likewise, the contingency is identified as an n-k contingency for 'k' outages. Conducting n-1 contingency analysis is more feasible in power systems compared to n-k contingency analysis, as n-1 failures are

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considerably more probable in real-world scenarios, while n - k events denote rare, simultaneous outages that are statistically less likely. Furthermore, industry standards, such as those established by the North American Electric Reliability Corporation (NERC), require compliance with n -1 contingency protocols as a foundational criterion for ensuring system reliability, whereas n - k analysis is not typically mandated for routine operations. Additionally, designing systems to accommodate all potential n - k contingencies would necessitate excessive redundancy and substantial infrastructure investments, rendering such approaches economically impractical. In instances where n - k analysis is warranted, such as in resilience assessments, it is generally confined to selected credible combinations through heuristic or probabilistic screening methodologies. Consequently, n -1 analysis remains the predominant and standard approach for power system planning and operation. The focus of this paper is on the evaluation of system performance parameters by the proposed methodology in estimating DA UC in the presence of n -1 thermal generator contingencies.

The dependable operation of power systems is of utmost importance to ensure an uninterrupted electricity supply, particularly as modern grids increasingly integrate renewable energy sources and face escalating demands. A significant challenge in maintaining grid stability involves addressing potential contingencies that may disrupt power generation. The N -1 contingency criteria, assuming the failure of a single generator or transmission line, are a widely adopted standard in power system planning and operation. It guarantees the system's functionality even when a critical component fails. Understanding the impact of these contingencies on the system's operational decisions is crucial for thermal power generators, which will continue to hold a substantial role in global power generation for some years till they are replaced by green energy [1].

UC, which determines the optimal scheduling of generation units to meet the forecasted demand at the lowest cost, is one of the primary decisions affected by N -1 contingencies. A robust UC schedule must account for the possibility of unforeseen outages in thermal generators while ensuring the availability of sufficient reserve capacity to uphold system reliability. However, thermal generator outages can give rise to significant operational challenges, including increased generation costs, grid instability, and the need for rapid adjustments in generation from alternative sources. Various studies on this topic propose different approaches to effectively carry out UC in the presence of n -1 contingencies. Numerous solutions have been proposed in previous research to address the issue of n -1 contingencies. One approach involves activating additional power plants to compensate for the outage of conventional generators. Specifically, gas power plants are employed in UC to ensure an uninterrupted power supply in the presence of n -1 generator contingencies [2]. Another option is to utilize the individual contributions or coordinated operation of interconnected or integrated system

components. For instance, a study investigates the coordinated operation of a combined heat & electricity system with renewable energy sources to enhance the flexibility of the power system in conducting energy dispatch under uncertain component states [3]. The modelling of uncertain outages of individual generators becomes necessary for developing a comprehensive probability distribution that proportionately affects UC scheduling and costs [4]. Thus, it holds significant importance in estimating DA UC costs in the event of generator contingencies.

Additionally, transmission line switching has been explored as a means to enhance system performance in optimal power flow problems when faced with n -1 contingencies [5]. Moreover, integrated renewable energy sources serve as alternative power sources to maintain system reliability and flexibility when determining optimal UC under n -1 contingencies [6]. While there has been research on n -1 generator contingencies, studies have also been conducted to investigate the impacts of n -1 transmission line contingencies on UC [7]. Power system security assessment for multiple contingencies has been empirically demonstrated in prior research [8]. However, most of these works focus on specific contingency management strategies or integrated systems, lacking a unified methodology to assess system-wide performance metrics—such as criticality, robustness, reliability, and operating margin—in the context of DA UC under N -1 generator outages.

The occurrence of n -1 generator contingencies has the potential to disrupt the supply-demand equilibrium, potentially leading to an inability to meet load demands. This situation necessitates a comprehensive analysis of system reliability, criticality, and the available operating margin to facilitate appropriate preparations for dispatch based on the DA UC results. This paper conducts an impact analysis on the reliability and robustness of DA UC decisions under n -1 thermal generator contingencies. By examining how the failure of individual thermal generators affects the optimal scheduling of generation units, our objective is to evaluate the grid's capacity to maintain supply-demand balance and meet load requirements during such challenging conditions.

The reliability of the system under these circumstances is assessed using metrics such as LOLP, Loss of Load Expectation (LOLE), etc. as established in previous studies [9]. Before estimating LOLP, probabilistic analyses are performed to evaluate the probabilities of generation unavailability under various n -1 generator contingencies [10], [11]. In this paper, we estimate the reliability of the n -1 generator contingency-based DA UC results using LOLP through the proposed methodology. Existing works that estimate LOLP generally do so for planning or risk analysis. However, very few integrate such probabilistic reliability indices within the context of DA UC execution under generator contingencies, which is necessary for system preparedness.

For the estimation of DA UC scheduling and associated costs, we employ Dynamic Programming (DP) optimization

in this study. In comparison to novel heuristic and metaheuristic approaches, DP optimization yields the most optimal solution [12]. Given the critical role of thermal generators in supplying base load and balancing intermittent renewable energy sources, this research provides valuable insights into how power system operators can formulate more resilient scheduling strategies for the following day in response to contingencies. By incorporating $n-1$ generator contingencies into the unit commitment planning process, power systems can enhance their preparedness to manage unforeseen outages during the execution of DA UC. Nevertheless, existing literature does not address the need for an explicit operating margin metric to evaluate the feasibility and headroom of DA UC decisions, particularly under $N-1$ generator outages. The severity of such outages can be quantified by developing measures that define and quantify system criticality, robustness, reliability, and operating margin concerning the DA UC problem for the forecasted hourly demand. Despite the wide range of contingency analysis tools and component-level criticality metrics, there is a clear absence of methodologies aimed at estimating and ranking the criticality of individual generators in DA UC scheduling under $N-1$ contingency events.

The power system criticality determination requires the assessment of system vulnerability, which can be performed by simulating various outage events and finding out the most critical components. The critical components of the power system have been analyzed in the Generation and Transmission sectors via the Failure Mode and Effects Analysis (FMEA). The early research in power systems involved the devising of criticality metrics for identifying critical components. One such study includes the development of metrics, Birnbaum, Fussell-Vesely, Reliability Achievement Worth (RAW), Reliability Reduction Worth (RRW), and Criticality Importance (CI) for assessing the criticality of the Transmission system [13]. Next to determining the criticality of components, the analysis of system criticality for various power system planning problems becomes an important area to work in. The analysis of Self-Organized Criticality (SOC) in power systems becomes necessary to further derive the resilience and adaptive strategies in the presence of outages. Mei et.al. performed SOC of the IEEE 118-bus system via the AC Optimal Power Flow method for analyzing system criticality in the presence of cascading failures and blackouts [14]. The critical states of the power system are assessed in [15] for a range of ambient temperatures, wind speeds, and demand volumes. The criticality index is assigned by estimating the distance from the limit surface. Another research study performed the ranking of critical assets via graph theory based on the simulated cascading failures for the integrated natural gas and electricity infrastructures [1]. The authors determined the criticality of the coupled natural gas and electricity infrastructures via the proposed geodesic vulnerability index in the presence of the simulated disruptions in their other study [2]. The system state estimation can also perform criticality analyses of power systems. The phasor measurements of the

power system obtained from Phasor Measurement Units are utilized in [16] to analyse the system's criticality. Criticality of the power system should also be determined when the system is subject to extreme and unpredictable natural events like earthquakes. One such research is performed by Espinoza et. al. via the risk and resilience estimation to assess the impact of seismic events [17]. While these works provide valuable insights into component and system-level criticality across various domains, the development of generator-specific criticality metrics directly influencing DA UC schedules remains a critical and unaddressed area.

While the criticality estimation for individual components and system are developed for various power system planning problems in smart grid [18], for individual/correlated outages [19] and for integrated systems [20]; the development of metrics for assessment of the criticality of individual generators and system criticality need to be worked upon for DA UC in the event of $n-1$ generator outages.

A power system possesses robustness if it can fulfil the load demand under adverse conditions like system disturbances, outages, and other extreme events. The recoverability of a normal power system operation makes it robust to the simulated events. Thus, robustness metrics define the extent to which the system can withstand such events. These have been developed for a variety of analytical problems like stability analyses, study of system dynamics, planning problems, etc. The early research on the development of frequency domain robustness margins and guidelines for their applications on Single Machine Infinite Bus (SMIB) Multi-Input Multi-Output (MIMO) control system representation of power systems was presented in 1984 [21]. Robustness of the system against cascading failures of p -fraction of transmission lines is simulated until the complete failure of the system takes place when all lines fail [22]. System robustness under cyber and physical attacks with line contingencies is assessed by load loss ratio for one bus attack and two lines attack based security constrained optimal power flow, which is optimized by combining particle swarm optimization and primal dual interior point methods [23]. The robustness of the power grid is enhanced in [24] by modifying the network topology. The authors developed topological metrics like average shortest path length, average degree of generators, average betweenness of generators etc. for finding out the impact of topological modifications on robustness. In [25], the functional robustness of power system dynamics in presence of faults was analyzed by the proposed metrics for three cases namely purely topological model, an artificial flow model, and a direct current power flow model. Robustness stability evaluation for SVC controllers was proposed in 1999 [26]. Many other studies are performed in the field with the development of many metrics. However, the robustness of DA UC when exposed to $n-1$ generator contingencies is not devised in the literature. Although numerous studies have developed robustness metrics for system stability, cascading failure analysis, and cyber-physical risks, robustness

quantification of DA UC plans under N-1 generator outages is yet to be formalized.

A. RESEARCH GAPS

Despite several advancements in UC and contingency analysis, the following significant research gaps persist:

- i) **Absence of an Integrated System Performance Framework:** Current research predominantly concentrates on discrete elements such as cost optimization, reserve scheduling, or generator coordination during contingencies. However, there is a conspicuous lack of a comprehensive methodology that simultaneously evaluates system criticality, robustness, reliability, and operating margin in the context of N-1 generator contingencies impacting DA UC.
- ii) **Lack of Established Metrics for Generator-Level Criticality in DA UC:** Criticality indices have been proposed for system components utilizing methodologies such as FMEA, SOC, or topological analysis. However, these frameworks are not specifically designed to assess the criticality of individual thermal generators in DA UC under N-1 contingency scenarios. A formalized mechanism to rank and quantify generator criticality is necessary to facilitate contingency-aware scheduling decisions.
- iii) **Inadequate Evaluation of Robustness for Contingency-Based DA UC Schedules:** Robustness metrics are typically applied to cascading failures, control systems, or network topologies. Nevertheless, the resilience of DA UC schedules in the context of generator outage scenarios has not been subjected to comprehensive analysis within the existing literature. This gap restricts the capacity to assess the effectiveness of these schedules in mitigating disturbances or sustaining operational feasibility during contingencies.
- iv) **Limited Incorporation of Probabilistic Reliability Metrics into DA UC Performance Evaluation:** Although probabilistic indices such as Loss of Load Probability (LOLP) and Loss of Load Expectation (LOLE) are well-established in power system reliability analyses, there is a scarcity of studies that directly integrate these measures into the evaluation of DA UC schedules. This integration is crucial for assessing the real-world viability of commitments exposed to contingencies.
- v) **Absence of a Quantified Operating Margin Metric for Dispatch Feasibility:** Most existing research on DA UC emphasizes the attainment of cost-minimizing schedules under predicted conditions, without providing a clear quantitative assessment of how closely the resultant schedule aligns with the system's operational limits. This limitation hampers operators' capacity to determine whether the schedule will remain feasible in the face of unforeseen contingencies.

In light of the aforementioned considerations, this study delineates several significant research gaps in the extant literature that impede the reliability and preparedness of DA

UC plans in the event of generator outages. These gaps encompass the absence of integrated system performance evaluations, generator-level criticality metrics, robustness analyses of UC schedules, probabilistic reliability assessments, and explicit operating margin estimations. Collectively, these deficiencies constitute the basis for the proposed methodology. Addressing these issues is crucial for improving real-time dispatch feasibility and enhancing long-term system resilience in the context of N-1 thermal generator contingencies.

B. CONTRIBUTIONS

This study presents a structured framework for evaluating system performance and preparedness for real-time dispatch in the presence of N-1 thermal generator contingencies. The proposed methodology introduces both quantitative and qualitative tools for assessing key system performance indicators—criticality, robustness, reliability, and operating margin—within a contingency-aware DA UC framework. The main contributions are as follows:

- i) A novel methodology for generator-level criticality assessment is developed to identify and quantify the operational impact of individual N-1 generator outages on DA UC schedules. This enables the classification and ranking of contingencies based on their severity, directly addressing the lack of generator-specific criticality metrics in existing literature.
- ii) A CM-based robustness evaluation is proposed to quantify the system's ability to sustain outages without violating operational limits. The CM metric supports the identification of weak and robust buses, addressing the current gap in robustness analysis of DA UC schedules under generator contingencies.
- iii) Integration of probabilistic reliability analysis using the LOLP index provides a quantitative assessment of system reliability across varying levels of Spinning Reserve (SR). This contributes to bridging the gap between reliability estimation and DA UC performance evaluation.
- iv) A new operating margin metric is formulated to determine the residual capacity available beyond committed reserves, reflecting the system's real-time feasibility and resilience. This explicitly addresses the need for quantifying dispatch margin under stress conditions, which is currently absent in conventional DA UC approaches.

In summary, this work addresses multiple gaps in existing literature by developing a comprehensive framework for evaluating system performance in contingency-aware DA UC. It introduces new metrics for assessing generator criticality, system robustness, reliability via LOLP, and operating margin—each of which is essential for reliable and economically feasible real-time dispatch planning under N-1 generator contingencies.

The structure of this paper is as follows: The proposed methodology for assessing the performance of n-1 contingency-based DA UC is discussed in Section II. The mathematical formulation of DA UC with the constraints,

including reliability assessment, is presented in Section III. Section IV provides an outline of the Case Studies employed to simulate the proposed n-1 contingency-based DA UC. The optimization technique employed for DA UC is mentioned in Section V. The following section presents the data for the IEEE RTS and the underlying assumptions for performing the proposed analysis. The results, along with their analysis, are presented in Section VII. Lastly, Section VIII offers conclusions.

II. PROPOSED METHODOLOGY

The proposed approach addresses key gaps in current literature by incorporating generator-level contingency impact assessment, robustness quantification, probabilistic reliability estimation, and operational margin evaluation into the DA UC framework. The methodology comprises the following steps.

Step 1: Identification of Critical n-1 Generator Contingencies

In the first step, N-1 contingency simulations are conducted for each thermal generator to identify outages that significantly impact DA UC feasibility or economic performance.

- (i) Contingency Impact Assessment: Generator outages are classified as critical if they cause operational infeasibility or substantial cost increases in the DA UC solution.
- (ii) DA UC Modeling: The DA UC problem is formulated and solved using DP for all contingency scenarios. The severity of each contingency is assessed based on the incurred operational cost or failure to meet the load. This directly addresses the lack of generator-specific criticality assessment methods in existing contingency-aware UC studies.

Step 2: Identification of Weak Buses Associated with Critical Contingencies

Upon identifying critical generator outages, the next step involves determining the weak buses, those most sensitive to associated failures. Weak buses are identified based on their susceptibility to overloads or operational stress during DA UC under critical contingencies. These buses are generally connected to or impacted by the failed generator units.

This step extends contingency analysis to the spatial dimension of the network, addressing the research gap in linking contingency severity to localized grid vulnerability.

Step 3: Estimation of System Robustness and Identification of Robust Buses

To quantify system robustness, this step identifies stable N-1 contingencies and the corresponding robust buses.

- i) Stable Contingency Classification: Generator outages that do not disrupt UC feasibility and/or do not increase cost significantly are deemed stable.
- ii) Robust Bus Identification: Buses connected to generators under stable contingencies are classified as robust. These nodes demonstrate the system's inherent ability

to absorb disturbances without compromising operational integrity.

This addresses the research gap regarding the absence of system-wide and bus-level robustness metrics for DA UC schedules.

Step 4: Reliability Estimation Using LOLP

This step integrates LOLP as a probabilistic metric to evaluate the reliability of DA UC outcomes under generator outages.

Based on probabilistic modeling of generator unavailability, the LOLP index is computed for each scenario. This reflects the likelihood that the system fails to meet demand under given contingencies and reserve conditions.

This contributes to the limited body of research that connects probabilistic reliability metrics with UC outcomes in a contingency-aware context.

Step 5: Evaluation of the Operating Margin

The final step involves the evaluation of the Operating Margin, a newly proposed metric that measures the system's additional capacity beyond committed reserves, capable of absorbing future contingencies.

The Operating Margin is calculated as the difference between the maximum allowable LOLP and the estimated LOLP value. A positive margin indicates headroom for additional load or contingencies; a negative margin implies risk of dispatch failure.

This metric directly supports system operators in assessing real-time dispatch feasibility, bridging the gap in quantifying flexibility and resilience of DA UC schedules under stress.

The proposed methodology is illustrated in Fig. 1.

III. PROBLEM FORMULATION

The n-1 generator contingency based hourly DA UC is performed after implementing the particular generator outage, to find out the minimum cost schedule of turning 'on' or 'off' of generators for the 24 hours of the next day, based on the constraints mentioned in Section III-B and the pre-determined load demand which is forecasted via the short term forecasting techniques.

A. OBJECTIVE FUNCTION

The objective function is the minimization of overall DA UC costs i.e. C_{Tot} . It comprises of aggregation of the fuel cost $F_{i,t}$, startup $StC_{i,t}$ and shut down costs $SdC_{i,t}$ of all committed generators for the 24 hours of the next day, given by eq. (1)

$$C_{Tot} = \sum_{i=1}^n \sum_{t=1}^T F_{i,t} \cdot u_{i,t} + StC_{i,t} \cdot u_{i,t} + SdC_{i,t} \cdot u_{i,t} \quad (1)$$

The commitment status is defined by $u_{i,t}$ which is '1' for a committed unit, and '0' for a decommitted unit for generator 'i' at hour 't'. The maintenance cost is neglected in the formulation.

Assuming DC power flow, the supply-demand balance is achieved if the overall generation by all the committed thermal generators is equal to the forecasted load demand at

hour 't' D_t^f as depicted in eq. (2)

$$\sum_{i=1}^n u_{i,t} \cdot P_{i,t} = \sum_{t=1}^T D_t^f \quad (2)$$

$P_{i,t}$ is the generated power of unit 'i' at hour 't'

B. CONSTRAINTS

The various constraints considered for estimating the DA UC costs are mentioned below from eq. (3) to (8), with others referred from [27] Generator Fuel Cost is the cost of fuel for running a generator 'i' for hour 't', where $\alpha_i, \beta_i, \gamma_i$ are the fuel cost coefficients

$$F_{i,t} = \alpha_i \cdot P_{i,t}^2 + \beta_i \cdot P_{i,t} + \gamma_i \quad (3)$$

Turbine Start-up Cost $StC_{i,t}^{Tbn}$, Boiler Start-up cost $StC_{i,t}^{Blr}$ and Maintenance Start-up cost $StC_{i,t}^M$ form the start-up cost $StC_{i,t}$ where hd and Cb are unit down hours and boiler cool down coefficient respectively.

$$StC_{i,t} = StC_{i,t}^{Tbn} + \left(1 - e^{\frac{hd}{Cb}}\right) \cdot StC_{i,t}^{Blr} + StC_{i,t}^M \quad (4)$$

$H_{d,i}$ is the number of hours a unit is down, $B_{c,i}$ is the Boiler Cool Down Coefficient.

The mathematical formulations of shutdown cost, minimum up time, minimum down time, active power limits on generating units, ramp up rate, and ramp down rate are referred from [27].

The ramp rate is a prevalent parameter in the field of power generation, serving as a measure of the rate at which power output varies with respect to time. Typically denoted in MW/min, it quantifies the speed of these changes specifically during ramping-up or ramping-down process.

The ramp-up rate defines the acceptable increase in power output at hour 't' with respect to the previous time interval 't-1' as per the maximum limit $P_i^{RU,max}$

$$(P_{i,t} - P_{i,(t-1)} \leq P_i^{RU,max}), \forall i, \forall t \quad (5)$$

The ramp down rate defines the acceptable decrease in power output at hour 't' with respect to the previous time interval 't-1' as per the minimum limit $P_i^{RD,min}$

$$(P_{i,t} - P_{i,(t-1)} \geq P_i^{RD,min}), \forall i, \forall t \quad (6)$$

Spinning Reserve: This is the 'generation reserve' which refers to the quantity of unutilized capacity in power generation that can serve as a back-up during instances of power shortages or generator failures. Typically, the generation reserve is maintained at a level equivalent to ten percent of the operating reserve.

$$R_t \geq (0.1 * R_i), \forall i, \forall t \quad (7)$$

In this work, with the simulation of n-1 generator contingencies R_t is utilized to improve system performance, by estimating CM as mentioned in eq. (9).

The total capacity of the system available for DA UC or load dispatch is obtained after subtracting this operating

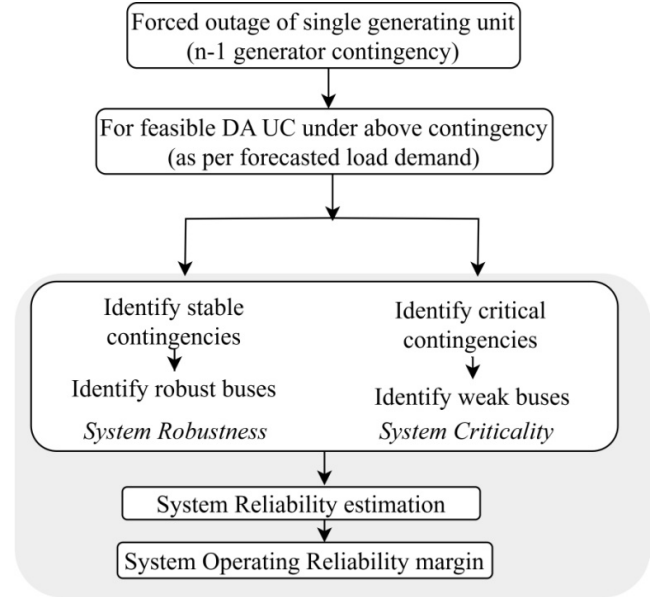


FIGURE 1. Proposed methodology for the system performance assessment of n-1 contingency based DA UC.

reserve which is not allocated in normal system operation rather utilized during contingency, forced outage etc. This available capacity is specified by eq. (8)

$$P^{avl} = \sum_{i=1}^n (P_i^{max} - R_i), \forall t \quad (8)$$

C. CONTINGENCY MARGIN

CM defines the reserve capacity that can be activated promptly to address unanticipated events [27]. For n-1 contingency based DA UC, CM in this paper is evaluated for contingency 'Cy' after consideration of the contributions from spinning reserve of all thermal generators, as expressed by eq. (9).

$$M_{da}^k = \sum_{i=1}^n R_i^k \quad (9)$$

R_i^{Cy} is the spinning reserve of generator 'i' for k^{th} contingency which is implemented as per the Case study defined in Section IV.

D. RELIABILITY ASSESSMENT via LOLP

The reliability of a system can be evaluated through the use of reliability indices. Various electricity regulatory authorities across different countries establish the maximum allowable limits for generator contingency reliability indices, such as LOLP and LOLE. A high probability of load loss indicates that generation is insufficient to meet demand, thereby suggesting a lower reliability of DA UC in the presence of the considered generator contingencies. Conversely, reliability is deemed higher when the probability of load loss remains below the established maximum allowable limit.

The maximum prescribed limit $LOLP_{max}$ for the k^{th} contingency $LOLP_k$ is defined by eq. (10)

$$LOLP_k \leq LOLP_{max}, k \in [1, Cy] \quad (10)$$

k belongs to the overall assessed contingencies from 1 to Cy , considering the various outages. Cy is referred to as the contingency index in this paper.

The system reliability is evaluated in this paper via LOLP. The reliability of the $n-1$ generator contingency-based DA UC is validated by the index LOLP. LOLP of each hour ' t ' of the next day, for ' k^{th} ' contingency simulated in the system, is given by eq. (11).

$$LOLP_{k,t} = Prob. \left[D_{f,t} > \sum_{i=1}^N (P_{i,t} \cdot u_{i,t}) \right], \forall t \quad (11)$$

The probability of load loss exists (i.e. non-zero) when the demand is greater than the overall available generation for that hour ' t '. The formulation and calculations for probability evaluation are referred to from [10] and [11]. The loss of load probability for an individual hour ' t ', for all possible $n-1$ generator contingencies as defined by contingency index Cy is evaluated by eq. (12).

$$LOLP_t = Prob. \left[D_{f,t} > \sum_{k=1}^{Cy} \sum_{i=1}^N (P_{k,i,t} \cdot u_{k,i,t}) \right], \forall t \quad (12)$$

LOLP defines the sum of probabilities of hourly forecasted demand being greater than the total power generation by thermal generators. For DA UC, this accounts for the sum of probabilities of the generation unavailability of 24 hours of the next day, which is represented in Eq. (13).

$$LOLP = \left(\sum_{t=1}^{24} LOLP_{k,t} \right) \quad (13)$$

The hourly LOLP, based on the considered contingencies in the system, will be non-zero only if there exist certain probabilities in an hour ' t ' where generation is not able to fulfill the forecasted load demand of that hour. If no such probabilities exist for the forecasted load demand of that hour, then LOLP will be zero. This is indicated by Eq. (14)

$$LOLP_{k,t} = \begin{cases} Prob. \left[D_{f,t} > \sum_{i=1}^N (P_{i,t} \cdot u_{i,t}) \right], & \text{if } P^{cap,avail} < D_{f,t}^t; \forall t \\ 0, & \text{otherwise}; \forall t \end{cases} \quad (14)$$

The hourly probability of loss of load for the next day are compiled in the Capacity Outage Probability Table (COPT).

E. OPERATING MARGIN EVALUATION OF THE SYSTEM

The operating margin of the system is evaluated by aggregating the load loss probabilities over the 24 hours of the subsequent day, as delineated in eq. (13). This assessment is conducted in the context of single generator contingencies to evaluate the performance of DA UC in relation to the

forecasted load demand in the event of a single generator outage. Furthermore, the operating margin of the system, which remains available in light of simulated $n-1$ generator contingencies for the day-ahead period, is estimated in this study using Eq. (15).

$$M_{da}^o = (LOLP_{max} - LOLP) \quad (15)$$

A positive operating margin M_{da}^o indicates a probability that the system can withstand the contingencies without experiencing a collapse. Conversely, a negative operating margin suggests a heightened probability of system failure under the conducted $n-1$ contingency analysis, which directly indicates that additional generation capacity is necessary to meet the demand.

F. ESTIMATING THE CRITICALITY AND ROBUSTNESS OF THE SYSTEM FOR $n-1$ GENERATOR CONTINGENCY BASED DA UC

The process of identifying system criticality points and system robustness is illustrated in Fig. 1. Throughout the paper, this process is employed to identify the weak and robust buses of the system in response to $n-1$ generator contingencies. If the DA UC is feasible based on the fulfillment of the supply-demand metric, the outcome may be either an economical or uneconomical DA UC. The stable contingencies, which result in an economical DA UC, can help identify the robust buses of the network. Conversely, if the DA UC is uneconomical, critical contingencies are identified to determine the weak buses.

IV. TECHNIQUE FOR DETERMINING $n-1$ GENERATOR CONTINGENCY BASED DA UC WITH VARYING CONTINGENCY MARGINS

This paper utilizes DP [28], [29] to determine the optimal solution for DA UC, specifically focusing on analyzing the costs associated with $n-1$ generator contingencies, referred to as $n-1$ generator contingency-based DA UC. Additionally, the effects of decreasing CM from ten percent to zero percent on the costs of $n-1$ generator contingency-based DA UC are analyzed to identify the optimal CM in the event of $n-1$ generator outages. The reliability of $n-1$ generator contingency-based DA UC results is estimated as per the formulation mentioned in the previous section. The economic performance of $n-1$ generator contingency-based DA UC is evaluated by comparing these results with those obtained without any contingency. Fig. 1 illustrates the process flow of the applied technique.

A. BASE CASE OR PRE-CONTINGENCY DA UC COST ANALYSIS WITHOUT $n-1$ GENERATOR CONTINGENCIES

To assess the economic performance of $n-1$ generator contingency-based DA UC, it is necessary to evaluate the UC costs for the system in the absence of any contingencies, referred to as the "Base Case" in this paper. The costs of the pre-contingency DA UC are then determined, forming the basis for the Base Case.

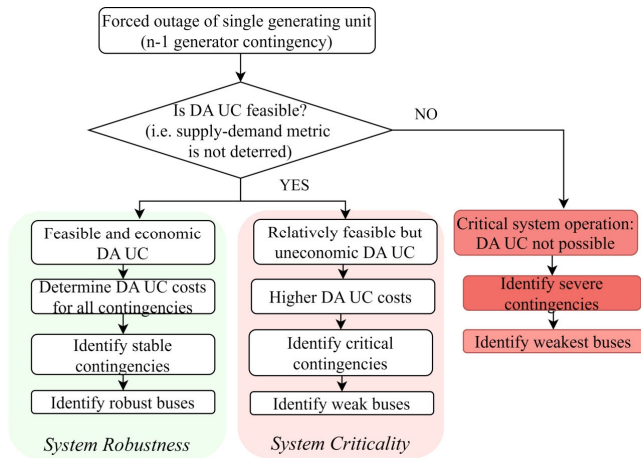


FIGURE 2. Estimation of criticality and robustness of system for n-1 generator contingency based DA UC.

B. DA UC COSTS FOR n-1 GENERATOR CONTINGENCIES WITH CM OPTIMIZATION-CASE 1 AND CASE 2

The n-1 generator contingency-based DA UC analysis involves the following steps, as explained below:

Case 1: Perform DA UC analysis for the various n-1 generator contingencies in the system

- Simulate the outage of 1 generator, one by one. For each generator contingency, run the DA UC with the mentioned constraints and CM corresponding to a ten percent SR. Compile the DA UC costs.
- Identify the stable and critical contingencies as per the proposed methodology (Fig. 2).
- Identify the corresponding robust and weak generator buses as per the proposed methodology (Fig. 2). Rank the weak buses in increasing order with the increasing value of percentage rise in UC costs. Rank the robust buses in the opposite sequence.

Case 2: Decrease the CM in steps from ten percent to zero percent and perform n-1 generator contingency-based DA UC analysis again to find out the optimal CM.

- Simulate the outage of 1 generator, one by one. For each generator contingency, run the DA UC with the mentioned constraints, decreasing values of SR from ten percent to zero percent. Compile the DA UC costs for the CM variation of 10%, 8%, 5%, and 0% with respect to the various n-1 generator contingencies.
- Perform steps (ii) and (iii) of Case 1.
- Identify which value of CM gives most economical DA UC when n-1 generator contingencies are simulated.

C. COMPILATION OF n-1 GENERATOR CONTINGENCY-BASED DA UC REPORT AND ANALYSIS

DA UC is performed on modified IEEE RTS [30], [31], [32], [33] using the DP technique. The results of the analysis described in Steps I and II of the previous section IV-B are compiled and analyzed. Thus the analysis is performed in the following three conditions:

- Base Case: The pre-contingency or 'Base Case' DA UC costs are obtained for the original IEEE RTS system, whose description and data are compiled in Section 5.1, and the underlying assumptions are mentioned in Section 5.2.
- Case 1: The n-1 generator contingency-based DA UC costs are evaluated with CM equal to ten percent
- Case 2: The n-1 generator contingency-based DA UC costs are evaluated with CM reduction in steps from 10% to 8%, 5%, and then 0%, respectively.

The above description is outlined in Table 1.

The DA UC costs of 'Base Case' are compared with the costs obtained after n-1 contingency analysis in Case 1 and 2. The contribution of CM (if available) in improving the DA UC costs is observed. The following are determined from the analysis:

- Ranking of buses to identify the weak buses for the n-1 generator contingency, based on the percentage rise in UC costs
- Optimal CM for DA UC: It is found by comparing the results of the DA UC costs for CM decrease from 10% to 0%. The contribution of CM decreases in the improvement of the weak generator bus ranking is observed.

V. METHODOLOGICAL FRAMEWORK FOR IMPLEMENTING THREE CASE STUDIES TO EVALUATE SYSTEM PERFORMANCE UNDER N-1 GENERATOR CONTINGENCY IN DA UC ANALYSIS

The framework for the implementation of the three case studies, as delineated in Table 1, seeks to evaluate system performance under simulated n-1 generator contingencies, as depicted in Figure 3. The analysis is conducted in two distinct stages. The first stage comprises the assessment of n-1 generator contingency-based DA UC, with its performance compared to the Base Case to enhance the understanding of the impacts of contingencies. The second stage entails a validation process wherein the results obtained from the generator contingency-based DA UC are corroborated through a load loss probability analysis, focusing exclusively on single generator contingencies. This is accomplished via the Loss of Load Probability evaluation for individual generator outages. Subsequently, the results of the LOLP analysis are utilized to estimate system reliability and the operating margin. The outcomes of the DA UC, in conjunction with the LOLP estimation and the evaluation of operating margins, serve as critical metrics for assessing system performance. This comprehensive evaluation aids in determining the feasibility of load dispatch for the forthcoming day, ensuring that the system remains within established operational limits while upholding the requisite level of reliability.

VI. SYSTEM DESCRIPTION

This work examines the use of the 24-bus, 26-generator IEEE Reliability Test System (RTS) [30], [31], [32], [33] for

performing n-1 generator contingency-based UC with optimal CM allocation. The single line diagram of IEEE RTS is shown in Fig. 4.

A. ASSUMPTIONS FOR CARRYING OUT THE PROPOSED ANALYSIS

The n-1 generator contingency is performed with the following assumptions:

- i) For n-1 contingency, failure of one unit is considered in the system with no simultaneous outages of single units of other rating generators. This is done since the simultaneous outage of two or more different rating generators is practically not possible to avoid shutdown & startup costs as well as maintenance. Moreover, the periodical maintenance of all generators in a running condition is ensured to avoid such forced outages.
- ii) The conditional probability of 'one' generator outage with other generators is not considered.

The evaluation of LOLP is based on the following assumptions:

- i) LOLP is assessed by considering the forced outage of only one generating unit at a time.
- ii) The paper does not take into account simultaneous outages of single units with different ratings connected to different bus numbers.
- iii) Outages of more than one unit are not considered for LOLP calculation.
- iv) Additionally, the conditional probabilities between various generators are not modeled to simplify the analysis. Furthermore, their impact on the results is considered negligible, and therefore, they are disregarded in the analysis.
- v) $LOLP_{max}$, as specified by the Central Electricity Authority (CEA) of India, is 0.05 [34]. This value can be verified from the official website data repository of the CEA. Similarly, the US Department of Energy specifies a $LOLP_{max}$ of 0.002 (equivalent to 0.1/year) [35], while the European Union Standards specify a $LOLP_{max}$ of 0.008 [36]. These values can be verified from the respective official website data repositories. For this study, assuming the system is operating under strict limits, a $LOLP_{max}$ of 0.05 is considered

B. SYSTEM DATA

The system comprises 26 thermal generators, which are assigned indexes ranging from 1 to 26, connected to a 24-bus system. The data from the IEEE RTS is cited in references [1], [2], [14], and [15]. The data utilized in this study are presented in Table 2. The bus numbers ('b') to which generators of specific capacity are connected are listed in column 1 of Table 2. The number of generating units connected to each bus number, their respective indexes, and the maximum capacity of a single unit are indicated in columns 2, 3, and 4, respectively. The final column provides the maximum generation capacity available at each bus 'b'.

TABLE 1. Cases for the performed analysis.

Case	Description
Base	DA UC cost determination for IEEE RTS at forecasted load, without contingencies
1	DA UC cost determination in the presence of all possible n-1 generator contingencies with CM of 10%
2	DA UC costs determination for the following three conditions with CM variation in steps from 8% to zero percent, incorporating all possible n-1 generator contingencies
	(a) 8% CM
	(b) 5% CM
	(c) Zero percent CM

The total available capacity of the system is 3105 MW. However, the capacity available for UC or load dispatch will be determined using Equations (1) to (9) with other constraint formulations referred from [27]. Therefore, considering the default 10% SR corresponding to CM of 310.5 MW (approximately 310 MW), the capacity available for estimating DA UC is **2795 MW**. This value will be used as a reference for the **Base Case and Case 1**.

In **Case 2**, the capacity available for dispatch can be summarized as follows:

Case 2(a): The 8% SR of each generator corresponds to CM of approximately 248 MW. With this CM, the capacity available for DA UC will be increased to **2857 MW**.

Case 2(b): The 5% SR of each generator renders CM of 155 MW unavailable to the operator for dispatch. As a result, the capacity available for DA UC becomes **2950 MW**.

Case 2(c): An SR of 0% i.e. zero CM indicates that all generators in the system are loaded to their full capacity. In this scenario, the capacity available for DA UC is **3105 MW**.

The generator data, including the fuel cost coefficients a_i , b_i , c_i , active power limits, ramp up rate, ramp down rate, hot & cold startup costs, cooling time constant, minimum up time, and minimum down time, are referred from [30], [31], and [32]. The mean time to failure (MTTF), with the failure rate (FR) of each unit, is mentioned in Table 3, as referred from [10] and [11]. FR, λ is related to MTTF i.e. t_{FR} as per eq. (16)

To facilitate the simulation of the n-1 thermal generator contingency-based DA UC, it is essential to first identify all potential contingencies arising from the various capacity generators installed across different bus numbers, as delineated in Tables 5 and 6. The identification and subsequent simulation of these generator contingencies will be conducted in accordance with the previously stated assumptions. The identified contingencies will be addressed utilizing the Contingency Index (Cy_k), as indicated by eq. 10. In this study, the Contingency Index Cy_k is assigned values ranging from 1 to 9, corresponding to the ascending order of bus numbers within the system network where generators are installed. A graphical representation of this assignment is illustrated in Fig. 5.

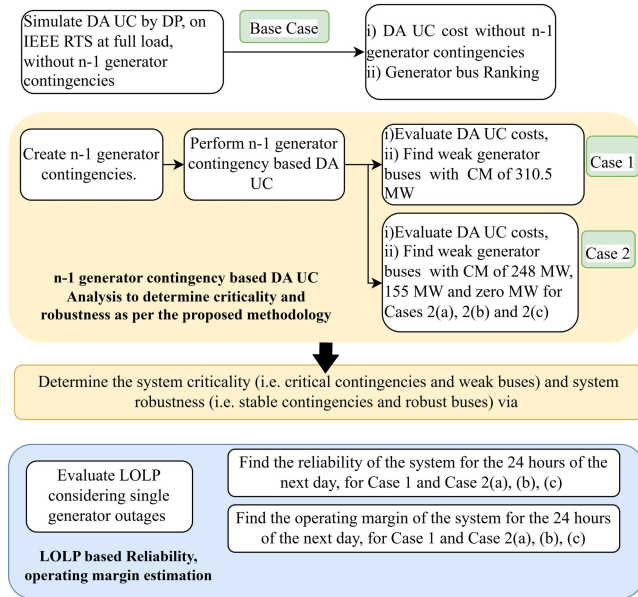


FIGURE 3. Methodological framework to evaluate system performance.

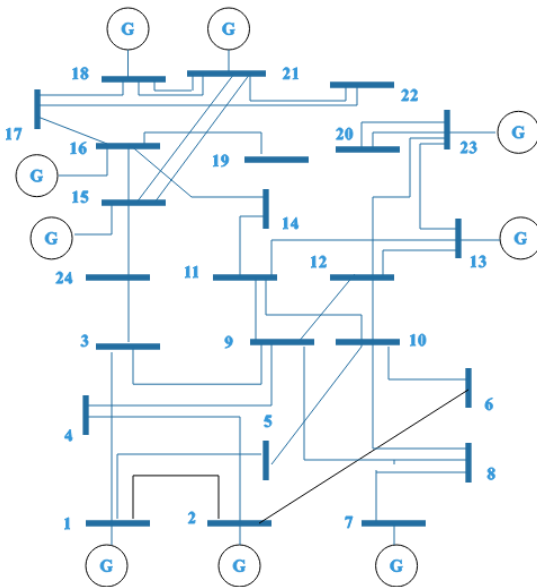


FIGURE 4. Single Line Diagram of IEEE RTS.

The anticipated load demand for the 24 hours of the following day is visually depicted in Fig. 6.

The expected highest demand on the next day will be 2670 MW at the 11th hour. Several other peaks can also be identified, which could potentially pose a risk when simulating all conceivable n-1 generator contingencies, especially if there is a shortage in the supply required to meet the demand due to generator outages.

VII. RESULTS AND DISCUSSION

This section presents the DA UC cost results for the Base Case, Case 1, and Case 2, followed by an evaluation of the system performance parameters as previously formulated

and discussed. Subsequently, a feasibility assessment is conducted regarding the load dispatch for the following day, informed by the system performance observed during the DA UC simulation under executed n-1 generator contingencies. The findings of the implemented methodology are detailed in Section VII-A. An analysis of these results is provided in Section VII-B, focusing on the practical application of DA UC for economically dispatching the loads for the following day and preparing the necessary measures in advance to ensure reliable system operation within safe limits.

A. COMPILATION OF DA UC RESULTS

After compiling the severe n-1 generator contingencies, the DA UC program is simulated with the outage of the one generator corresponding to each severe contingency. To estimate the increase in costs (if any) for each n-1 generator contingency, the pre-contingency or 'Base Case' UC costs are compared with these costs. The DA UC costs are estimated for the following cases:

- i) Base Case
- ii) Case 1
- iii) Case 2

This is followed by reliability analysis of the n-1 contingency-based DA UC results via LOLP.

1) BASE CASE

In the 'Base Case', the optimization of DA UC is performed via the DP technique with the mentioned constraints, when n-1 generator contingencies are not considered in the system, with a CM of 10%. Since the analysis is performed before simulating contingencies, it is termed as 'pre-contingency' analysis which forms the basis of comparison for the other 2 cases. The original system of IEEE RTS is utilized for conducting the analysis. The cumulative DA UC costs for each hour 't' of the next day are estimated for the system via DP. DP is executed in MATLAB via coding, including all the mentioned constraints, and compiled in Table 4.

$$\lambda = \frac{1}{t_{FR}} \quad (16)$$

2) CASE 1

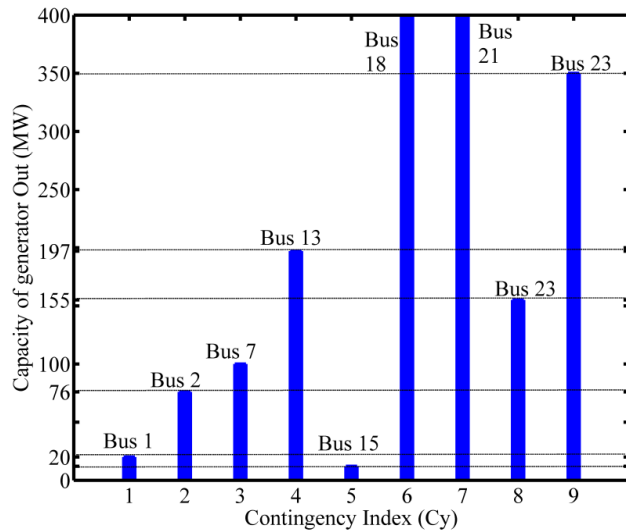
The impact of n-1 generator contingencies on the real-time UC and load dispatch is assessed by performing the forced outage of each generator as indicated in Table 5, for the specific contingency index mentioned in Column 1. The bus number from which the generator is removed for simulating n-1 contingency is mentioned in Column 2. The capacity of the generator, which is not available in the system for generation, is mentioned in Column 3. The fourth column shows the DA UC cost of the Base Case. The DA UC costs with 10% SR due to the outage of the corresponding generator are mentioned in Column 5. The percentage increase in the

TABLE 2. Data of generation units.

Bus No. (b)	Thermal generating units		Max. Capacity of single generating unit (MW)	Max. available Capacity connected to bus 'b'
	No. of units	Index of Units		
15	5	1,2,3,4,5	12	12x5=60
1	4	6,7,8,9	20	20x4=80
2	4	10,11,12,13	76	76x4=304
7	3	14,15,16	100	100x3=300
23	4	17,18,19,20	155	155x4=620
13	3	21,22,23	197	197x3=591
23	1	24	350	350
18,21	2	25,26	400	400x2=800
	Total Units=26			Total Available Generation capacity = 3105

TABLE 3. Mean time to failure and failure rate of different capacity thermal generators.

Bus No. (b)	Thermal unit max. capacity (MW)	MTTF (t_{FR})	FR (λ)
15	12	2940	0.0003401
1	20	450	0.00222
2	76	1960	0.0005102
7	100	1200	0.00083
23	155	960	0.00104
13	197	950	0.00105
23	350	1150	0.00086956
18,21	400	1100	0.0009

**FIGURE 5.** Identified 9 n-1 thermal generator contingencies for IEEE RTS.

DA UC costs for the various n-1 contingencies is mentioned in Column 6.

The assignment of bus rank is determined by calculating the increase in DA UC costs during the simulation of the specific n-1 contingency. The higher the decrease in the percentage rise of UC costs, the higher the assigned rank. This information can be found in the last column. Upon examination, it is evident that Buses 18 and 21 are the weakest

buses under the n-1 generator contingency for DA UC since the costs are 15.67% higher. Generator outage at bus number 23 results in an 11.67% and 3.67% rise in DA UC costs for the 350 MW and 155 MW generator contingency, respectively. Bus 23 is the next weak bus, thus assigned the rank 2. A very slight increase in DA UC costs is observed for the outage of 76 MW and 100 MW generators at buses 2 and 7, respectively. Therefore, these are ranked 3 and 4 in the same sequence.

The outage of one 20 MW unit, at bus 1, doesn't affect the DA UC costs. Additionally, the outage of one generator at bus 13 (197 MW) and bus 15 (12 MW) results in a slight decrease in costs compared to the Base Case. Hence, the buses 1, 13, and 15 don't witness any uneconomical effect of n-1 generator contingency based on the DA UC.

3) CASE 2

In Case 1, CM was 10%. In this case, SR is reduced to 8%, 5%, and 0%, consequently, for assessing the impacts of CM utilization during the specific contingency with the index mentioned in Column 1. For each of these values of CM, the n-1 generator contingency-based DA UC analysis is performed again to find out the overall DA UC costs as compiled in Table 6. The UC cost for the Base Case is mentioned in column 4. Column 5 presents the UC costs of Case 1, i.e., with 10% SR. The UC costs of Case 2 for the various CM levels are shown in the last three columns.

It can be observed from the table that the decrease of CM from 10% to 8%, and then from 8% to 5%, doesn't have any significant impact on DA UC costs despite the decrease in costs for n-1 contingency at bus 13 and bus 15. However, when CM is further reduced from 5% to 0%, the decrease in DA UC costs is notable for most of the simulated contingencies, as highlighted in the last column of Table 6.

B. RELIABILITY ESTIMATION OF n-1 GENERATOR CONTINGENCY BASED DA UC

The reliability of the Case 1 and Case 2 results is estimated via the reliability index LOLP. For estimating the reliability of the obtained DA UC results under n-1 generator contingency, LOLP is calculated. To calculate the LOLP index, the generation unavailability needs to be evaluated from the perspective of the various n-1 generator contingencies for different cases of 'one' generator outages. The table containing this data is termed as Capacity Outage Probability Table (COPT), as shown in Tables 7 and 8 for Cases 1 and 2, respectively. The probability of generation unavailability is calculated in terms of FR and repair rate from the mathematical formulation referred to in [10] and [11].

1) COPT CALCULATIONS FOR CASES 1 AND 2

Tables 7 and 8 represent COPT for Cases 1 and 2. The probability of generation unavailability for the single unit outage is obtained from the FR mentioned in Eq. (15) after considering the assumptions mentioned in Section 5.2.

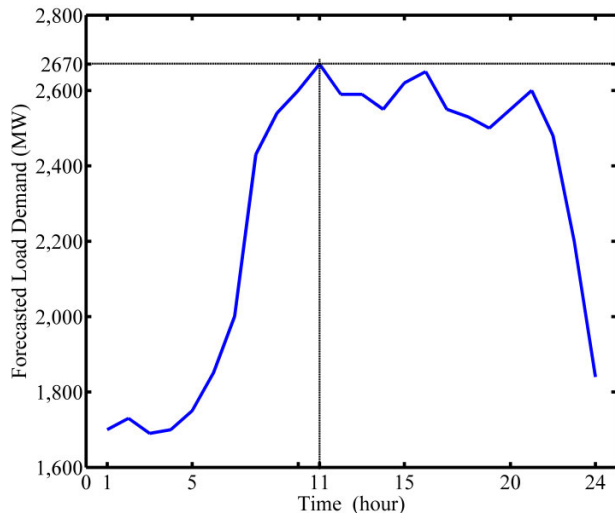


FIGURE 6. Hourly Forecasted load demand of IEEE RTS [9].

TABLE 4. DA hourly cumulative UC costs for 'Base Case'.

Hour (t)	UC cost C_{uc} (\$)	Hour (t)	UC cost C_{uc} (\$)
1	20086	13	380268
2	40623	14	418051
3	60560	15	457446
4	80646	16	497571
5	101487	17	535354
6	123882	18	572713
7	149731	19	609444
8	185836	20	647227
9	223406	21	686137
10	262316	22	722457
11	302931	23	751863
12	341600	24	774100

Case 1: The total capacity available for UC allocation after simulating the outage of a single generating unit is mentioned in column 3 of Table 7. The last column of Table 7 compiles the probability of generation unavailability for each n-1 generator contingency. The available capacity for dispatch is 2795 MW as calculated in Section VI-B. The highest value of forecasted load demand is 2670 MW at hour '11', as can be observed from Table 4. From Table 7, the total capacity for DA UC doesn't reduce lower than the highest value of demand, i.e., 2670 MW, for the n-1 contingencies on bus numbers 1, 2, 7, and 13. Thus, the probability of generation unavailability can occur only for the n-1 contingencies occurring at bus numbers 15, 18, 21, and 23.

Case 2: The available capacity for dispatch is 2857 MW, 2950 MW, and 3105 MW, respectively, for Case 2 (a), (b), and (c) as calculated in Section VI-B. The total capacities available after the outage and their probabilities are compiled in Table 8 for three considerations of Case 2, which are discussed as under:

- From Table 8, the total capacity for DA UC doesn't reduce lower than the highest value of demand, i.e., 2670 MW, for the n-1 contingencies on bus numbers 1,

2, 7, 15, and 23 (for the outage of a 155 MW generator). Thus, the probability of generation unavailability can occur only for the n-1 contingencies occurring at bus numbers 13, 18, 21, and 23 (for the outage of the 350 MW generator).

- The total available capacity after generator outage is lower than the highest demand at bus numbers 18, 21, and 23 (for an outage of a 350 MW generator), thus having a high probability of loss of load during generator outage.
- All buses carry sufficient generation capacity even after the loss of one generator. Hence in this case, the probability of loss of load during generator outage becomes zero.

2) LOLP CALCULATIONS FOR CASES 1 AND 2

The loss of load probability of individual hour, for all identified contingencies (referred to as C_y in this paper) is evaluated from Eq. (12) and compiled in Table 9. These are calculated by aggregating the individual single generator outage probabilities as mentioned in Tables 9 and 10 for Cases 1 and 2, respectively.

The overall LOLP for the 24 hours of the next day is determined from Eq. (13).

Case 1: The probability of loss of load for individual hours with the overall probability of load loss for the next day is compiled in Table 9. For the considered system, the estimated value of overall LOLP for the next day is 0.050113, as can be observed from Table 9.

Case 2: The probability of loss of load for individual hours is compiled in Table 10 for Case 2(a), (b), and (c). The estimated value of overall LOLP for the next day is **0.03693587**, **0.0153346**, and **0** for Case 2(a), 2(b), and 2(c), respectively, as can be observed from Table 10.

C. OPERATING MARGIN

The operating margin is evaluated from Eq. (15) for Cases 1 and 2. By referring to the maximum limit of loss of load probability considered in this study (i.e., 0.05), and the obtained results of LOLP for the simulated Case studies as compiled in Tables 9 and 10, the obtained value of operating margin M_{da}^o are calculated and compiled in Table 11.

D. ANALYSIS OF RESULTS

The results obtained are judged and analyzed utilizing the aforementioned data. The analyses derived from the simulated cases are derived and summarized in this section.

1) COMPARATIVE ANALYSIS OF D.A. U.C. COSTS FOR THE N-1 GENERATOR CONTINGENCIES WITH CRITICALITY AND ROBUSTNESS ANALYSIS OF SYSTEM

a: BASE CASE

The DA UC cost for the considered IEEE RTS system when optimized via DP is found to be \$ 774100. This cost forms the basis of comparison for the n-1 contingency-based DA UC with varying SR, which are analyzed in Cases 1 and 2.

TABLE 5. Comparison of Case 1 with Base Case: DA UC costs for n-1 generator contingencies with 10% CM.

Contingency Index (Cy_k)	Bus No.	Capacity of Generator out (MW)	Base Case UC cost (\$)	Case 1 Cost (\$)	Percentage increase in cost (%)	Ranking
1	1	20	774100	774100	0	-
2	2	76		779635	0.715	3
3	7	100		778409	0.557	4
4	13	197		768660	-0.703	-
5	15	12		774020	-0.01	-
6	18	400		895400	15.67	1
7	21	400		895400	15.67	1
8	23	155		797762	3.06	2
9	23	350		864440	11.67	2

TABLE 6. DA UC costs for n-1 generator contingencies: Comparison of Base Case, Case 1 and Case 2.

Contingency Index (Cy_k)	Bus No.	Capacity of Generator out (MW)	Base Case UC cost (\$)	Case 1 UC costs (10% CM) (\$)	Case 2 UC costs (\$)		
					(a)	(b)	(c)
1	1	20	774100	774100	774100	774100	773109
2	2	76		779635	779635	779635	777438
3	7	100		778409	778409	778409	775201
4	13	197		768660	768660	768660	768660
5	15	12		774020	774020	774020	773069
6	18	400		895400	895400	895400	895400
7	21	400		895400	895400	895400	895400
8	23	155		797762	797762	797762	791763
9	23	350		864440	864440	864440	864440

TABLE 7. Case 1: Capacity outage probability Table (COPT) for the outage of a single generating unit.

Contingency Index (Cy_k)	Bus number (b)	Max Capacity of outage (MW)	Total Capacity available after generator outage (MW)	Probability of Generation Unavailability after a Generator Outage
1	1	20	2783	0.00222
2	2	76	2775	0.0005102
3	7	100	2719	0.000833
4	13	197	2695	0.001052
5	15	12	2640	0.0003401
6	18	400	2508	0.000909
7	21	400	2508	0.000909
8	23	155	2455	0.001041
9	23	350	2395	0.00086956

The Base Case cost of DA UC is quite low, i.e., without any generator outage, the system runs smoothly.

b: CASE 1

Even with a 10% SR, it is still possible to perform DA UC based on n-1 generator outages. These contingencies do not pose a significant threat to the supply-demand balance, and real-time load dispatch can still be effectively carried out

despite the large increase in DA UC costs in the case of one generator outage on buses 18, 21, and 23. N-1 generator contingency on bus numbers 2 and 7 results in a negligible increase in UC costs. The remaining buses don't experience any increase in costs despite the generator outage.

The percentage variation of DA UC costs obtained in Case 1 with respect to Base Case results is represented in Table 5. The percentage variation in DA UC costs for Cases 2(a) and 2(b) is also the same. A graphical representation of the percentage variation of these costs with respect to the nine simulated contingencies is shown in Fig. 7.

After determining the percentage variation in UC costs for contingencies indexed from 1 to 9, they are subsequently ranked from 1 to 4 based on the descending order of their DA UC costs. These rankings are graphically presented in Fig. 8.

Fig. 8 illustrates the ranking of nine simulated n-1 generator contingencies, specifically for Case 1 and the two conditions of Case 2- (a) and (b), when CM is 248 MW and 155 MW, respectively, as illustrated in Section VI-B.

Case 1 and Case 2(a), 2(b) are analyzed together due to the identical DA UC costs observed between Case 1 and the aforementioned conditions of Case 2, as confirmed by the data presented in Table 6. It is evident that the highest capacity generator outage, specifically 400 MW at buses 18 and 21, poses the greatest risk to the system. Thus, **buses 18 and 21 are the weakest** in the network. Consequently, **contingencies (Cy)6 and 7** are ranked as the **most critical**, occupying the top position. Contingency 1, on the other hand, does not exhibit any increase in DA UC costs, thereby posing no threat to the system. **Contingencies 4 and 5** demonstrate a decrease in costs below the Base Case UC cost, making them the **most stable** contingencies. Thus, **bus 13 and 15 serve as the robust buses in the network**.

Thus, the n-1 generator contingency, even with 10% SR, doesn't have any significant impact on DA UC implementation since the supply-demand balance is maintained. The only change observed is the increase in DA UC costs for a few contingencies. The above analysis is done for 10% SR with a CM of 310.5 MW. However, to utilize CM in a better manner for further reducing DA UC costs in the presence of n-1 generator contingencies, CM is reduced in steps from 248 MW to 0 MW in Case 2.

c: CASE 2

The post-contingency costs are quite high for the high-capacity generators of 350 MW and 400 MW. For the generators of 300 MW and 450 MW, their outage leads to high UC costs, since the other higher cost generators need to be committed in their place to deliver power to the required load demand. However, with the full operating reserve consumption during the n-1 contingency, i.e., when CM is 0 MW, the UC costs drop significantly compared to those of Case 1.

The percentage variation in DA UC costs for Case 2 needs to be analyzed for the following three conditions, as specified in Section VI-B:

TABLE 8. Case 2: Capacity Outage Probability Table (COPT) for outage of single generating unit.

Contingency Index (Cy_k)	Bus number (b)	Max Capacity of outage (MW)	Case 2(a)		Case 2(b)		Case 2(c)	
			Total Capacity available after generator outage (MW)	Probability of Generation Unavailability after generator outage	Total Capacity available after generator outage (MW)	Probability of Generation Unavailability after generator outage	Total Capacity available after generator outage (MW)	Probability of Generation Unavailability after generator outage
1	1	20	2837	0.00222	2930	0.00222	3085	0.00222
2	2	76	2781	0.0005102	2874	0.0005102	3029	0.0005102
3	7	100	2757	0.000833	2850	0.000833	3005	0.000833
4	13	197	2660	0.001052	2753	0.001052	2908	0.001052
5	15	12	2845	0.0003401	2938	0.0003401	3093	0.0003401
6	18	400	2457	0.000909	2550	0.000909	2705	0.000909
7	21	400	2457	0.000909	2550	0.000909	2705	0.000909
8	23	155	2702	0.001041	2795	0.001041	2950	0.001041
9	23	350	2507	0.00086956	2600	0.00086956	2755	0.00086956

- CM of 248 MW corresponding to an aggregated reserve of 10%
- CM of 155 MW corresponding to an aggregated reserve of 8%
- CM of zero MW percent for 0% SR

The results for the above-mentioned conditions of Case 2 are presented in Table 6. The DA UC costs for conditions (a) and (b) are the same as those of Case 1. Therefore, the determination of percentage variation in DA UC costs and ranking of contingencies will be the same as discussed in the previous sub-section for Case 1, and represented in Fig. 7 and 8. The critical contingencies and the weak buses are the same. Similarly, the stable contingencies and the robust buses are also the same.

However, the DA UC costs for condition (c) are different from those of Case 1 as well as Case 2(a) and 2(b), as can be observed from Table 6. Therefore, the assessment will be derived from the percentage variation in DA UC costs as plotted in Fig. 9, and the contingency ranking, which is shown graphically in Fig. 10.

It is evident from Fig. 9 that the DA UC costs are the highest for contingencies 6,7 in Case 2(c). Contingency 9 also shows the next sharp rise in UC costs. The percentage rise in UC costs is very low for contingency 8, followed by contingency 2, and is lowest for contingency 3. On the other hand, the DA UC costs reduce for contingencies 1, 4, and 5. Comparing the percentage variation in DA UC costs for Case 1 and Case 2(c) from Fig. 5 and 7, the DA UC costs reduce in Case 2(a) for the contingencies 1, 2, 3, 4, 5, and 8.

The rank of contingency is decided based on the economic DA UC results. The ranking of buses for Case 2(c) can be interpreted from the graphical presentation shown in Fig. 10. A rank of 1 corresponds to the most expensive/uneconomic contingency, and a rank of 5 shows the comparative least cost option. It can be observed that the outage of the largest generating unit of the system, i.e., 400 MW connected at buses 18 and 21, increases the system risk. Contingencies 6,7

TABLE 9. Case 1: LOLP (for forced outage of single generating unit).

Hour, t	Forecasted load (MW)	LOLP _t	Hour, t	Forecasted load (MW)	LOLP _t
1	1700	0	13	2590	0.0037285
2	1730	0	14	2550	0.0037285
3	1690	0	15	2620	0.0037285
4	1700	0	16	2650	0.00406866
5	1750	0	17	2550	0.0037285
6	1850	0	18	2530	0.0037285
7	2000	0	19	2500	0.00191056
8	2430	0.00086956	20	2550	0.0037285
9	2540	0.0037285	21	2600	0.0037285
10	2600	0.0037285	22	2480	0.00191056
11	2670	0.00406866	23	2200	0
12	2590	0.0037285	24	1840	0
LOLP					0.050113

occupy rank 1, with contingency 3 ranked last. Buses **18, 21** are therefore identified as the **weakest buses** of the network, with the corresponding contingencies **6 and 7** posing as the **most critical contingencies** of all. Contingencies 1, 4, and 5 exhibit cost reduction with costs lower than those in the Base Case. **Contingency 4** observes the lowest DA UC cost of all, lower than that of the Base Case, making it the **most stable contingency**, with **bus 13** as the **robust bus** of the system. The **next robust buses** in sequence are **buses 1 and 15**, corresponding to the **contingencies 1 and 5**.

While optimizing the CM utilization leads to an economical DA UC solution, the costs of Case 2(c) are still higher than the Base Case but lower than Case 1.

The following can be inferred from Case 2:

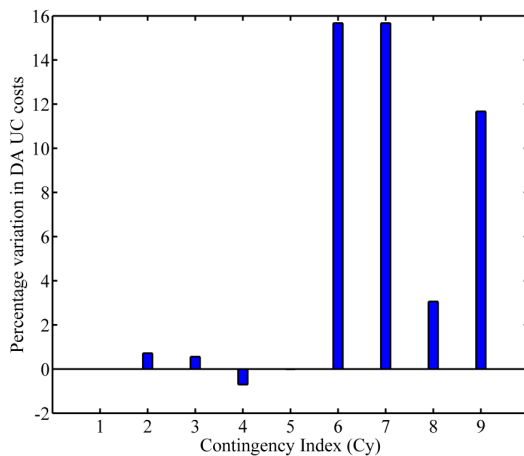
- With 8% aggregated SR corresponding to a CM of 248 MW, the system operator can dispatch an additional 248 MW compared to the Base Case and Case 1.
- The CM of 155 MW means there is an additional capacity of 155 MW compared to Base Case and Case 1, which can be utilized to serve consumers.

TABLE 10. Case 2: LOLP (for forced outage of single generating unit).

t	D_t^f (MW)	LOLP _t			t	D_t^f (MW)	LOLP _t		
		2(a)	2(b)	2(c)			2(a)	2(b)	2(c)
1	1700	0	0	0	13	2590	0.002687	0.001818	0
2	1730	0	0	0	14	2550	0.002687	0	0
3	1690	0	0	0	15	2620	0.002687	0.002687	0
4	1700	0	0	0	16	2650	0.002687	0.002687	0
5	1750	0	0	0	17	2550	0.002687	0	0
6	1850	0	0	0	18	2530	0.002687	0	0
7	2000	0	0	0	19	2500	0.001818	0	0
8	2430	0	0	0	20	2550	0.002687	0	0
9	2540	0.002687	0	0	21	2600	0.002687	0.001818	0
10	2600	0.002687	0.001818	0	22	2480	0.001818	0	0
11	2670	0.003739	0.002687	0	23	2200	0	0	0
12	2590	0.002687	0.001818	0	24	1840	0	0	0
LOLP							0.0369387	0.01533	0

TABLE 11. Operating margin for Cases 1 and 2.

Case	Case 1	Case 2		
		2(a)	2(b)	2(c)
Operating Margin	-0.000113	0.0130613	0.0346654	0.05

**FIGURE 7.** Percentage variation in DA UC costs for Case 1 and Case 2(a), 2(b).

- c) For zero CM, the available dispatch capacity is 310 MW more than the Base Case and Case 1. However, it is important to note that this situation poses a higher risk of cascading failures, collapse, and other critical failures since the system doesn't have additional capacity to mitigate these disturbances, apart from withstanding the simulated contingencies. Therefore, it is not utilized in normal system operations, but rather reserved only for contingencies or extreme emergency conditions.
- d) Thus, the most economical CM is that corresponding to 0% aggregated SR, indicating that generators are operated at their maximum capacity. However, due to

the aforementioned drawbacks, this approach is only employed in such exceptional circumstances.

2) RELIABILITY ASSESSMENT

The reliability assessment of the system under the condition of n-1 generator contingencies is conducted using the LOLP metric. The compiled results, as presented in Tables 9 and 10, demonstrate the following for the considered Cases.

Case 1: The overall LOLP for the DA UC is **0.050113**, which is more than the limit of 0.05. This suggests a lower reliability and indicates the requirement for additional generation to meet the load demand.

Case 2(a): The aggregated LOLP for the next day for n-1 generator contingency based DA UC is **0.0369387**, which is lower than the maximum limit, showing comparatively better reliability than Case 1.

Case 2(b): The DA UC LOLP is estimated as **0.0153346**, which is lower than that of Case 2(a), further improving the reliability.

Case 2(c): The **zero** value of load loss probability shows the highest reliability of DA UC under the simulated contingencies.

This finding confirms that the reliability of DA UC is adversely affected by the n-1 generator contingencies.

3) OPERATING MARGIN OF THE SYSTEM

By referencing the maximum LOLP threshold established in this study (i.e., 0.05), the assessed values of the operating reliability margin, denoted as M_{da}^o , for both Case 1 and Case 2 are derived using eq. (15).

In Case 1, a negative margin of -0.000113 signifies that the generators are operating at their limits, thereby necessitating an increase in system generation to meet demand under n-1 generator contingency scenarios. This increase can be achieved by utilizing the existing 10% SR associated with a CM of 310.5 MW.

In Case 2(a), a positive margin of 0.0130613, albeit minimal, indicates that the DA UC can be executed without risking system collapse. The system demonstrates the ability to withstand minor disturbances; however, it remains inadequate to endure additional contingencies or larger-scale disruptions. The available CM of 248 MW can be employed to mitigate such disturbances and enhance system performance.

In Case 2(b), the margin of 0.0346654 reflects a slight improvement compared to those observed in Case 1 and Case 2(a), suggesting an enhanced operational capability of the system to withstand disturbances. Nonetheless, CM decreases in this scenario to 155 MW, indicating a reduced ability of the system to endure further disturbances.

In Case 2(c), the margin of 0.05 is attained by depleting the system's CM to zero. This indicates that, while the system can operate effectively under certain contingencies, it becomes vulnerable to failure should any further contingencies or disturbances arise, as the capacity margin has been exhausted. Consequently, any additional stress on the system, such as simultaneous contingencies, higher-order contingencies (e.g.,

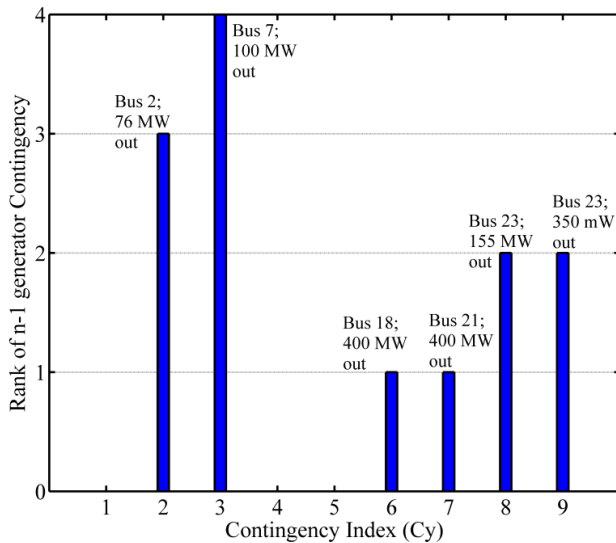


FIGURE 8. Ranking of n-1 generator contingencies for Case 1 and Case 2(a), 2(b).

n-2, n-k, etc.), or unexpected loading conditions, may precipitate a system collapse.

4) FEASIBILITY ASSESSMENT OF LOAD DISPATCH BASED ON THE SYSTEM PERFORMANCE

The performance of the system for DA UC under the nine n-1 generator contingencies, as outlined by the proposed methodology, can be evaluated through the assessment of criticality, robustness, reliability, and operating margin.

This analysis is structured into two distinct stages. In the initial stage, the criticality and robustness of the system are derived from the DA UC results corresponding to the simulated generator contingencies, as detailed in Table 12.

The second stage involves an evaluation of the reliability of the analysis conducted, alongside the assessment of the available operating margin, which is summarized in Table 13.

The comprehensive evaluation of the system's performance for DA UC under n-1 generator contingencies is presented in Tables 12 and 13. The feasibility of real-time load dispatch is examined based on the system's performance under the simulated n-1 generator contingencies. Following this analysis, corrective actions aimed at enhancing the system performance of DA UC under the nine contingencies are formulated.

VIII. DISCUSSION AND FUTURE WORK

This discussion confirms that system preparedness in DA UC is not solely a function of optimal cost but also of robust topology, reliability indices, and residual operating margins. The framework and findings presented in this work offer a valuable foundation for enhancing contingency-aware operational planning in modern power systems.

A. DISCUSSION

This study presents a comprehensive framework for evaluating power system performance under N-1 generator

contingencies in DA UC. The proposed methodology integrates four critical metrics—system criticality, robustness, reliability, and operating margin—to assess both the technical and operational feasibility of real-time dispatch decisions based on DA UC outcomes.

1) QUANTITATIVE ANALYSIS

The numerical results (Table 12) confirm the significance of accounting for generator-level contingencies. Despite maintaining the same lowest DA UC cost of \$774,020, the highest cost escalated to \$895,400 under critical contingencies across all cases, underscoring the economic impact of generator outages. The system exhibited a consistently high vulnerability to Contingency Cy6 and Cy7, with associated weak buses (18 and 21), indicating that these nodes are particularly susceptible to operational disruptions.

Conversely, robust buses (13, 15, and occasionally 1) were identified under stable contingencies, signifying regions of the grid that can absorb generator outages with minimal impact. The CM progressively diminished from 310.5 MW in Case 1 to 0 MW in Case 2(c), as spinning reserves increased from 0% to 10%, highlighting the trade-off between reserve allocation and system flexibility.

Reliability, quantified using LOLP, improved substantially from 0.050113 in Case 1 to 0.0 in Case 2(c) (Table 13). It illustrates that reserve provisioning has a direct and measurable impact on system reliability. Accordingly, the operating margin improved from a negative value of -0.000113 to a maximum of 0.05, suggesting enhanced real-time dispatch feasibility and resilience.

Furthermore, the observed relationship between increasing SRs and decreasing CM levels highlights a key trade-off in operational planning. In line with this, practical selection of CM thresholds can be guided by several operational criteria to enhance real-world applicability of the proposed framework:

- i) Forecast Uncertainty: CM can be dynamically adjusted based on expected variability in load or renewable generation forecasts. In systems with high renewable penetration, this flexibility becomes particularly valuable for maintaining stability under uncertain conditions.
- ii) Cost–Risk Trade-offs: While a higher CM improves contingency resilience, it also increases operating costs due to larger reserve commitments. Hence, selecting CM levels involves balancing economic efficiency against reliability goals—an essential consideration for dispatch planners.
- iii) Historical Contingency Impact: Past records of contingency frequency and severity can inform acceptable CM thresholds, helping operators determine risk tolerance based on system behavior trends.
- iv) Regulatory Reserve Requirements: CM planning can be aligned with minimum reserve or SR mandates imposed by system operators or regulatory agencies, ensuring compliance while maintaining flexibility.

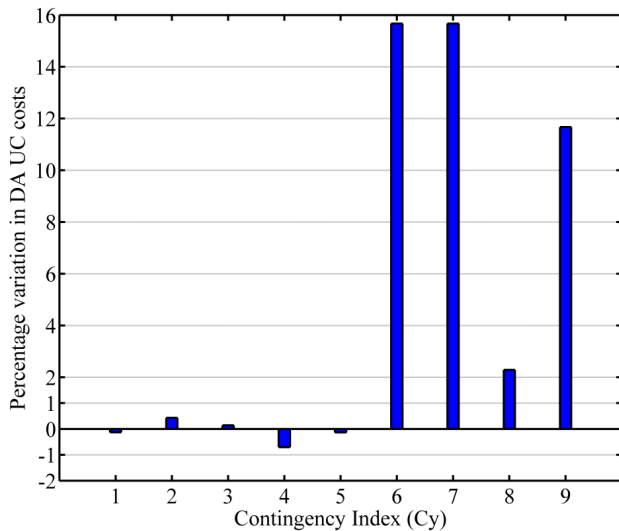


FIGURE 9. Percentage variation in DA UC costs for Case 2(c).

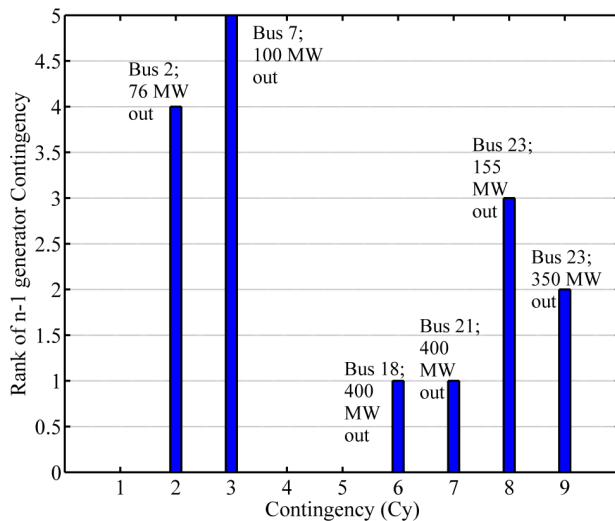


FIGURE 10. Ranking of n-1 generator contingencies for Case 2(c).

- v) Real-Time System Conditions: Adaptive CM adjustment may also be implemented closer to real-time, taking into account current generator availability, network congestion, and overall system stress levels.

This study fills key gaps in the literature by:

- i) Providing generator-specific criticality ranking in DA UC
- ii) Quantifying bus-level vulnerabilities and strengths under outages
- iii) Integrating probabilistic reliability assessment via LOLP
- iv) Introducing a practical operating margin metric for dispatch feasibility.

Unlike prior works focused solely on optimization or outage mitigation, this framework allows system operators to interpret how prepared a DA UC schedule is for real-time

dispatch when contingencies occur. This enables proactive decision-making and aids in identifying target corrective actions at weak buses.

2) QUALITATIVE ANALYSIS

In addition to the quantitative metrics, the qualitative insights derived from the proposed methodology offer significant value for operational planning.

The classification of generator contingencies into critical and stable events, along with the identification of corresponding weak and robust buses, enables operators to develop location-specific strategies for corrective actions. For instance, buses identified as weak (e.g., Bus 18 and Bus 21) in response to critical contingencies (Cy6 and Cy7) highlight spatial vulnerabilities in the system topology that may require reinforcement through targeted generation placement or network reconfiguration.

The robustness evaluation further differentiates between system areas with high inherent resilience and those that may suffer from cascading failures under outage scenarios. Moreover, the integration of LOLP and operating margin provides a system-level interpretability framework, where a negative margin directly correlates to an elevated risk of dispatch infeasibility.

These qualitative evaluations contribute to a deeper understanding of system behavior, extending beyond cost metrics to inform strategic investments, preventive scheduling, and emergency response planning in real-time grid operations.

B. SCOPE OF FUTURE WORK

While the proposed methodology facilitates a comprehensive performance evaluation, future integration of advanced techniques are expected to enhance performance and resilience. Recent advances in wildfire-resilient unit commitment modeling [38], real-time LSTM-based congestion control [39], and infeasibility-aware optimization [40] provide a foundation for extending the current framework. Moreover, quantum and distributed optimization paradigms [41] offer novel opportunities for scalability and security in large-scale power system scheduling.

The current study centers on thermal generator contingencies. Subsequent research could integrate renewable generation uncertainties and distributed energy resource (DER) integration to assess the response of hybrid systems to contingencies.

Future investigations may consider the incorporation of new and recent metaheuristic techniques—such as Particle Swarm Optimization (PSO), Grey Wolf Optimizer (GWO), Ant Lion Optimizer (ALO), or newly proposed algorithms such as the Rabbit and Turtle Algorithm [42], Farmer and Seasons Algorithm [43], and Negotiators Algorithm [44]—to evaluate trade-offs in solution quality, computational efficiency, and scalability.

The proposed framework operates under the assumption of a day-ahead planning horizon. Future extensions could

TABLE 12. Assessment of system performance for DA UC under n-1 generator contingency.

System Performance Parameters		Case 1	Case 2		
			2(a)	2(b)	2(c)
DA UC cost (\$)	Lowest cost of operation under contingency	774020	774020	774020	774020
	Highest cost of operation under contingency	895400	895400	895400	895400
System Criticality	Critical Contingencies (Cy)	6, 7	6, 7	6, 7	6, 7
	Corresponding weak buses	18, 21	18, 21	18, 21	18, 21
System Robustness	Stable Contingencies	4, 5	4, 5	4, 5	4, 1, 5
	Corresponding Robust buses	13, 15	13, 15	13, 15	13, 1, 15
DA UC performance under the nine simulated n-1 generator contingencies	SR committed for generation during contingency	0%	2%	5%	10% (full)
	CM remaining (to be utilized for generation)	310.5 MW	248 MW	155 MW	0
	System Operation in case critical contingencies arise	High cost operation	High cost operation	High cost operation	High cost operation with fully loaded generators
	Withstanding capability for any further contingency/ loading/ other related events	Poor	Lower	Low	Comparatively better
	Possibility of failure under contingency	High	Comparatively low	Further lower	Lowest
	Measures (within system range) to improve performance	CM needs to be utilized			Additional measures

TABLE 13. Assessment of feasibility of real time load dispatch and corrective actions.

System Performance Parameters		Case 1	Case 2		
			2(a)	2(b)	2(c)
Reliability assessment	Max. allowable limit of LOLP	0.05	0.05	0.05	0.05
	Evaluated LOLP	0.050113	0.0369387	0.0153346	0
	Reliability	Very low	Higher Compared to Case 1	Higher Compared to 2(a)	Highest Compared to all
Operating margin, M_{da}^o	Estimated value	-0.000113	0.0130613	0.0346654	0.05
	Does margin exist?	No	Yes	Yes	Yes
	Description	Heightened probability of system failure if generation is not augmented	Low withstand capability	Increased withstand capability	Adequate withstand capability
System Operation	In the context of reliability	Working on extreme limits	Limited margin	Increased margin	Reliable operation
Real-time dispatch	Feasibility of load dispatch based on DA UC results	Not possible for longer hours	Possible	Possible	Possible
Corrective Actions	Measures to enhance system performance	Alternate generation installation, preferably at identified critical buses.			

encompass real-time corrective scheduling mechanisms, including dynamic re-commitment or demand response, potentially leveraging deep learning-based forecasting and control models such as LSTM architectures [39] to address emerging generator or line failures.

Analysis of multi-region or interconnected grid scenarios, as well as systems featuring integrated gas-electric or heat-electric coordination, may provide insights into cross-domain robustness and criticality during coordinated outages. Recent work using consensus-based quantum decomposition

techniques [41] could also be investigated to enable scalable and secure scheduling in interconnected grids.

The proposed metrics could be incorporated into a visual analytics or software platform that offers real-time alerts and contingency impact dashboards, thereby aiding grid operators in critical decision-making.

Further research could extend the methodology to accommodate multi-contingency (N-k) scenarios and renewable generation uncertainties, potentially adopting stochastic or dynamic programming approaches [37], [38] for robust scenario modeling.

Additionally, investigating the application of the proposed framework to larger and more complex power systems would be beneficial.

IX. CONCLUSION

This methodology aims to assess and optimize the performance of a power system by examining its criticalities, robustness, and reliability, particularly under contingencies that could impact the DA UC process. The identification of critical n-1 contingencies, the analysis of weak and robust buses, the use of the LOLP index for reliability, and the evaluation of operating margins provide a comprehensive framework for ensuring both economic and operational efficiency of DA UC under n-1 generator contingencies, while maintaining grid stability.

A positive operating margin indicates a greater likelihood that the system can endure contingencies without experiencing failure. In contrast, a negative operating margin signifies an increased risk of system collapse, as demonstrated by the conducted n-k contingency analysis. This finding directly suggests that additional generation capacity is essential to meet demand.

A negative operating margin arises when DA UC is not feasible due to an imbalance between supply and demand, resulting in significantly reduced generation capacity. In such instances, if the generation deficit cannot be mitigated through alternative sources, renewable integration, storage system installation, or other implemented contingency measures, the system is at risk of failure. Conversely, if the negative operating margin associated with DA UC can be addressed, the identified weakest buses should be augmented with available measures following regional electricity regulations.

In conclusion, the integration of criticality assessment, robustness evaluation, and reliability estimation offers a powerful framework for improving the efficiency and stability of power systems under contingency conditions. The insights gained from this methodology are essential for grid operators and planners to design systems that are not only cost-effective but also resilient to failures. By considering the DA UC assessment as per the proposed methodology, power system operators can make informed decisions to enhance system security, optimize unit commitment strategies, and ensure a reliable and robust power supply for consumers while economically dispatching the load. In addition, the incorporation

of emerging metaheuristics can be explored in future studies to improve optimization performance in complex unit commitment scenarios.

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